1 Numerical experiments

We evaluate SFRAMES on spectrally constrained low-rank reconstruction from real low-dose computed tomography (CT) measurements. The experiments examine how sampling angular views affects progress at a fixed projection budget, and compare SFRAMES with SPRING–SARAH, ProxSARAH and PALM. Increasing the number of training views enlarges the finite sum while preserving the factor dimensions, allowing us to compare minibatch and full-batch updates under both measurement densities. Evaluation uses six 2DeteCT slices, with $N \in \{360, 1{,}080\}$ training views and 360 held-out views per slice.

1.1 Spectrally constrained CT reconstruction

An attenuation image describes how strongly material reduces transmitted X-ray intensity per unit distance. At each angular position, or view, the source illuminates a line of detector bins along a fan of rays. Under the Beer–Lambert model, the negative logarithm of normalized transmitted intensity approximates the line integral of attenuation along a ray [1]. We represent attenuation by an image $X \in \mathbb{R}^{n\times n}$, with $n = 512$. Let $\ell_{ij,pq}$ be the length in millimeters of ray j of view i inside pixel (p, q). The forward map for that view is

$$P_i \colon \mathbb{R}^{n\times n} \longrightarrow \mathbb{R}^d, \qquad [P_i(X)]_j = \sum_{p,q} \ell_{ij,pq} X_{pq}, \qquad d = 956. \tag{1}$$

The scanner geometry fixes the weights. The adjoint P_i^*, called backprojection, distributes detector residuals over the crossed pixels using the same weights.

We fit factors $U, V \in \mathbb{R}^{n\times r}$, with $r = 64$, so that $X = UV^\top$ has rank at most r. For processed measurements $y_i \in \mathbb{R}^d$ and a training-view set $\mathcal{I}$ of size N, define

$$f(U,V) = \frac{1}{N}\sum_{i\in\mathcal{I}} q_i(U,V), \qquad q_i(U,V) = \frac{\|P_i(UV^\top) - y_i\|_2^2}{2d}. \tag{2}$$

We bound each factor's largest singular value and impose entrywise nonnegativity through

$$\begin{aligned} C &= \{(U,V) : \|U\|_{\mathrm{op}} \le R_U,\ \|V\|_{\mathrm{op}} \le R_V\}, \\ D &= \{(U,V) : U \ge 0,\ V \ge 0\}. \end{aligned} \tag{3}$$

With $x = (U, V)$, $T = \mathrm{Id}$ and $g = \iota_D$, the problem is

$$\min_{x\in C,\, Tx\in D} f(x) \ = \ \min_{x\in C}\{f(x) + g(Tx)\}. \tag{4}$$

Here ι_D is zero on D and $+\infty$ elsewhere. Nonnegative factors ensure a nonnegative reconstructed attenuation image. The set C is compact and convex, and strictly positive sufficiently small factors lie in the interiors of both C and D. Each component has a Lipschitz gradient on a neighborhood of C. A smooth compactly supported extension equal to the component on that neighborhood gives a globally Lipschitz gradient while preserving the values and gradients at every queried factor pair.

For a uniform subset $B \subset \mathcal{I}$, a joint stochastic gradient is

$$E_B = \frac{1}{|B|d}\sum_{i\in B} P_i^*(P_i(UV^\top) - y_i), \qquad \widehat{\nabla} f_B(U,V) = (E_B V, E_B^\top U). \tag{5}$$

It is conditionally unbiased when B is drawn independently of the current factors. Each selected view contributes all d detector bins. One invocation uses a forward projection and an adjoint over the selected views. The full-batch gradient uses $B = \mathcal{I}$.

The orthant projection is $\Pi_D(x) = x_+$, applied entrywise to both factors. Thus the smoothed term is $g_\beta(x) = \mathrm{dist}_D(x)^2/(2\beta)$, with gradient $(x - \Pi_D(x))/\beta$. The implemented SFRAMES direction and update are

$$s_k = \mathrm{LMO}_C\left(m_k + \frac{x_k - \Pi_D(x_k)}{\beta_k}\right), \qquad x_{k+1} = (1-\gamma_k)x_k + \gamma_k s_k, \tag{6}$$

where $m_0 = \widehat{\nabla} f_{B_0}(x_0)$ and $m_k = (1-\alpha_k)m_{k-1} + \alpha_k \widehat{\nabla} f_{B_k}(x_k)$ for $k \geq 1$. For a block direction $G = A\,\mathrm{diag}(\sigma)B^\top$, the spectral-ball LMO returns $-RA_+B_+^\top$, with singular vectors restricted to positive singular values. The two factors are handled independently, using one thin SVD per factor.

Initialization and SFRAMES parameters. All methods start from common nonnegative rank-64 factors U_0, V_0, obtained by an NNDSVD-style factorization of a sensitivity-normalized backprojection of 120 training views, followed by a small positive perturbation and a scale fit. Set $R_U = 2\|U_0\|_{\mathrm{op}}$ and $R_V = 2\|V_0\|_{\mathrm{op}}$. Ten power iterations on the stacked calibration operator P_J estimate $\widehat{\lambda} \approx \|P_J\|_{\mathrm{op}}^2/(120d)$. We scale the smoothing prefactor by

$$L_U = \widehat{\lambda}\|V_0\|_{\mathrm{op}}^2, \qquad L_V = \widehat{\lambda}\|U_0\|_{\mathrm{op}}^2, \qquad \beta_0 = c_\beta/\max(L_U, L_V).$$

These initial block-curvature estimates provide a heuristic search scale. The joint development searches select $(c_\beta, \gamma_0) = (16, 0.1)$ for both SFRAMES variants. The selection criterion is repaired training loss relative to its initial value. Minibatch SFRAMES draws 64 training views uniformly without replacement at each update, independently across updates. Full-batch SFRAMES uses all N views. Both use

$$\gamma_k = \gamma_0(k+1)^{-3/4}, \qquad \alpha_k = (k+1)^{-1/2}, \qquad \beta_k = \beta_0(k+1)^{-1/4}.$$

Evaluation criteria. Raw training loss is $f(U, V)$. To evaluate a feasible reconstruction, we repair each factor by

$$\mathcal{R}_R(A) = \frac{A_+}{\max\{1, \|A_+\|_{\mathrm{op}}/R\}}, \qquad \widetilde{X} = \mathcal{R}_{R_U}(U)\mathcal{R}_{R_V}(V)^\top.$$

Repaired training loss uses $\widetilde{X}$ in (2). Held-out loss uses the same normalization over the 360 reserved views. Relative image error is $\|\widetilde{X} - X_{\mathrm{ref}}\|_F/\|X_{\mathrm{ref}}\|_F$, where X_{ref} is the aligned high-dose reconstruction. Feasibility is measured by the orthant distance

$$d_D = \left(\|\min(U, 0)\|_F^2 + \|\min(V, 0)\|_F^2\right)^{1/2}.$$

The repair is used only for diagnostics.

1.2 Results

The comparison contains 132 complete evaluation runs, with a common budget of 17,280 forward and 17,280 adjoint views. Table 1 reports reconstruction quality and feasibility. Relative to full-batch SFRAMES, the median paired change in repaired training loss is -12.7% for $N = 360$ and -34.4% for $N = 1{,}080$. Relative to SPRING–SARAH, the median paired change in repaired training loss is -32.0% for $N = 360$ and -35.9% for $N = 1{,}080$. Relative to ProxSARAH, the median paired change in repaired training loss is -29.4% for $N = 360$ and -28.1% for $N = 1{,}080$. Relative to PALM, the median paired change in repaired training loss is -43.3% for $N = 360$ and -55.8% for $N = 1{,}080$. Negative changes favor minibatch SFRAMES. For every comparator and training-view count, all six slices favor minibatch SFRAMES in median paired repaired loss. Paired percentages use the comparator as denominator and can differ from ratios of the marginal medians in the table.

Figures 1 and 2 show the observed losses and orthant distances against iterations, gradient-oracle calls and processed views. Both SFRAMES variants leave D after their first update and retain nonzero

Table 1: Reconstruction quality at the saved within-budget iterates. Entries take medians over seeds within each slice, then across six slices. Raw and repaired are training losses. Held-out loss and relative reference error use repaired factors. Lower is better. Bold entries give the best value within each training-view count, including ties.

N	Method	Raw	Repaired	Held-out	Ref. error	d_D
360	Minibatch SFRAMES	**0.01068**	**0.01085**	**0.01141**	**0.3326**	0.05044
360	Full-batch SFRAMES	0.01200	0.01238	0.01258	0.3715	0.09464
360	SPRING–SARAH	0.01613	0.01613	0.01615	0.4904	**0**
360	ProxSARAH	0.01543	0.01543	0.01547	0.4751	**0**
360	PALM	0.01909	0.01909	0.01909	0.5397	**0**
1,080	Minibatch SFRAMES	**0.01092**	**0.01105**	**0.01127**	**0.2888**	0.04937
1,080	Full-batch SFRAMES	0.01476	0.01670	0.01673	0.4668	0.2492
1,080	SPRING–SARAH	0.01735	0.01735	0.01734	0.5116	**0**
1,080	ProxSARAH	0.01543	0.01543	0.01543	0.4737	**0**
1,080	PALM	0.02362	0.02362	0.02358	0.6165	**0**

orthant distance at the work cap. Across the SFRAMES runs, the largest absolute raw–repaired endpoint-loss difference is 16.89% of the repaired loss. The projection baselines remain in D throughout.

These descriptive comparisons use six slices from two specimen tubes. The baseline step multipliers are retained from their development searches. Both selected SFRAMES pairs are interior to their final development grids. The observed comparisons apply to the stated parameter grids and data budget.

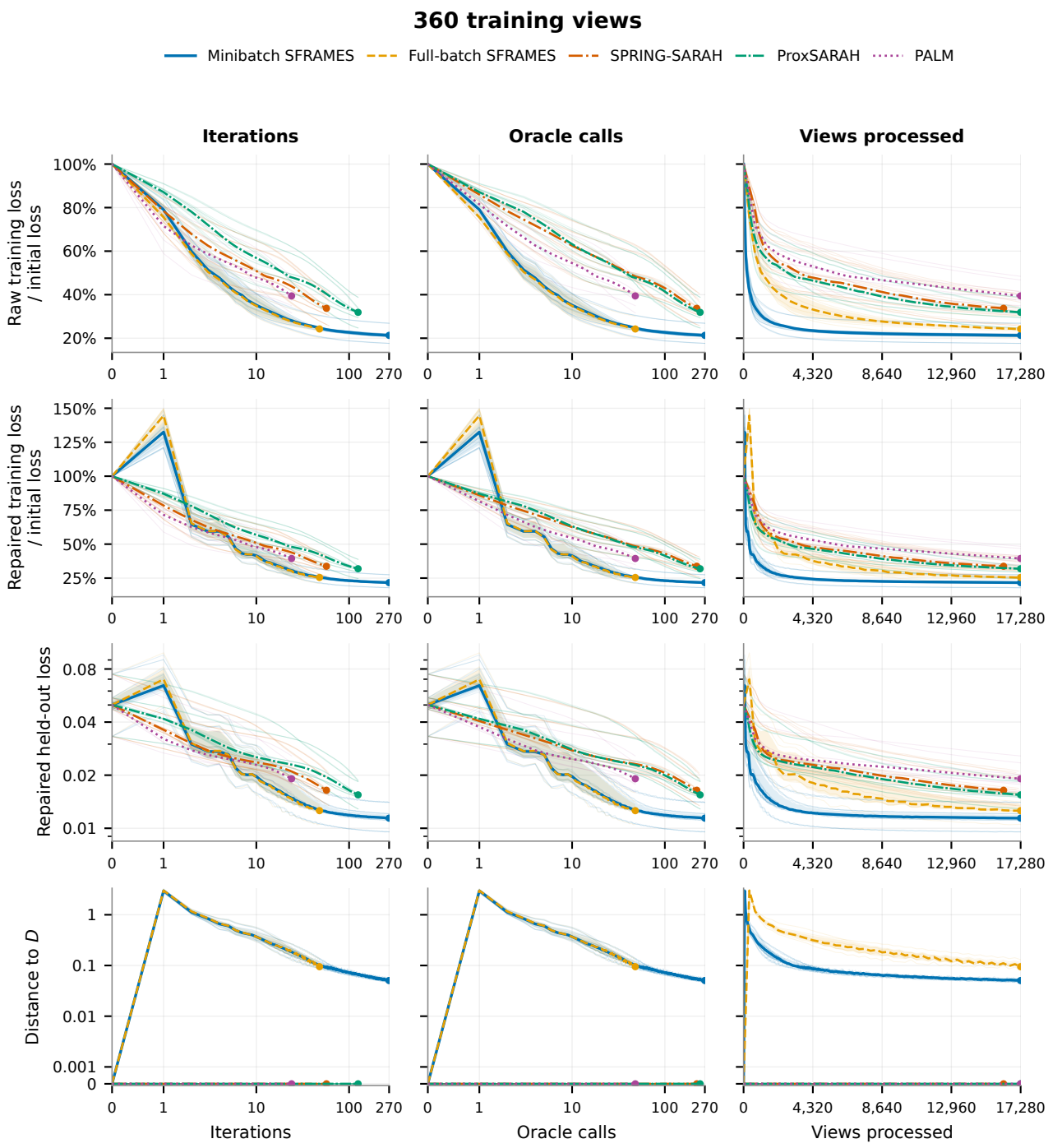


Figure 1: Trajectories with 360 training views. Thin lines show individual runs, recorded at every update. Bold lines take median work and metric at the same iteration, first over seeds within slices and then across six slices. Bands show the middle 50% of slice medians for SFRAMES. Training losses are normalized by their initial values.

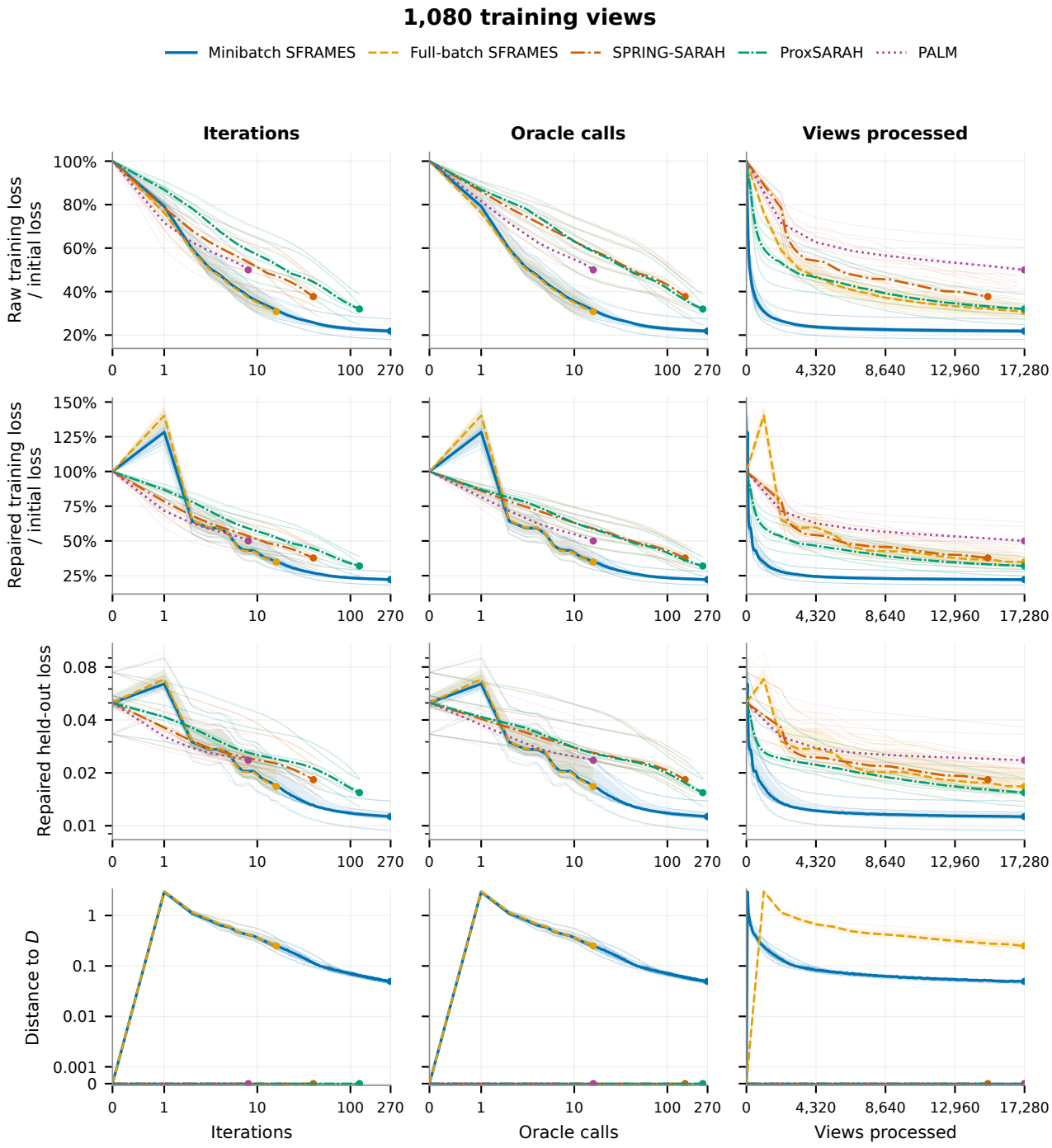


Figure 2: Trajectories with 1,080 training views, using the same metrics and summaries as Figure 1. The three projection baselines coincide at zero orthant distance.

1.3 Experimental protocol

We use measured fan-beam acquisitions from 2DeteCT [1]. Mode 1 supplies the low-dose measurements, and the published Mode 2 reconstruction from a separate high-dose acquisition supplies the image reference. Development uses slices `04300` and `03150`. Evaluation uses `03041`, `03381`, and `03611` from Mix 2, and `03791`, `04131`, and `04931` from Mix 3. Each group contains cross sections of one specimen tube.

We discard the repeated final angle, sum adjacent detector cells, subtract the similarly binned dark field, and divide by the mean flat field after dark correction. Taking the negative logarithm with a transmission floor of 10^{-6}, then reversing the detector axis, gives $3{,}600 \times 956$ sinograms in the DeepInverse convention [2]. The floor is inactive for all selected acquisitions. ASTRA 2.5.0 applies the FP32 CPU `line_fanflat` projector [3], while the factor arithmetic is FP64. The source-to-origin and origin-to-detector distances are 431.019989 and 97.980499 mm, with detector spacing 0.1496 mm. The square field has width 116.528143 mm and center $(-0.113797, -0.113797)$ mm, and view i has angle $\pi - 2\pi i/3600$. We vertically flip the 1024×1024 reference, divide by its native voxel width 0.113797 mm, and average 2×2 blocks.

For $i \in \{0, \dots, 3599\}$, the training sets select $i \bmod 10 = 0$ for $N = 360$ and $i \bmod 10 \in \{0, 3, 6\}$ for $N = 1{,}080$. Both reserve the 360 views with $i \bmod 10 = 9$ for evaluation. Calibration uses the 120 training views with $i \bmod 30 = 0$, shared across both sizes. The stochastic methods use three paired oracle seed labels $202, 203, 204$, with generator seeds offset by 100,000. Development uses label 110. PALM and full-batch SFRAMES run once per slice and training size and serve as shared deterministic references.

All methods start from the same calibrated factors in $C \cap D$. Writing P_J for the calibration operator, we divide $(P_J^* y_J)_+$ entrywise by $P_J^* P_J \mathbf{1}$, with denominator floor 10^{-3} times its maximum. A nonnegative SVD-based factorization gives rank-64 factors, perturbed by 10^{-3} times each factor's RMS times uniform $[0, 1)$ entries. We fit a least-squares amplitude on the calibration views with lower bound 10^{-6}, applying its square root to both factors. Calibration uses seed 110 in development and 201 in evaluation. The numerical LMO retains singular values above $\varepsilon_{\mathrm{mach}} \max(\operatorname{shape} G) \sigma_{\max}(G)$.

SPRING–SARAH [4] alternates factor updates with steps $a/L_U, a/L_V$. Each block independently refreshes with probability $1/q$, starting from a full gradient, and otherwise uses a paired 64-view SARAH difference. ProxSARAH [5] uses joint steps $a/(L_U + L_V)$, 256-view snapshots followed by m recursive 64-view updates, and relaxation 0.8. Both terms of each recursive difference use the same sampled views. PALM [6] uses full gradients with alternating steps $a/L_U, a/L_V$. Proximal steps project each factor onto its spectral ball intersected with the nonnegative orthant. A feasible single-set projection is accepted directly. Otherwise Dykstra iterations use a primal–dual certificate, requested distance tolerance $10^{-8}(k+1)^{-1.1}$, at most 2,000 iterations, and checks at the first, every fifth, and final iteration, with a floating-point allowance.

Table 2: Development hyperparameters. The two SFRAMES rows list complete Cartesian grids for the joint smoothing multiplier and step prefactor, with the selected pair in the final column. Baseline settings are retained from their original development searches.

Method and parameter	Values tested	Selected
Minibatch SFRAMES (c_β, γ_0)	$c_\beta \in \{0.25, 1, 4, 16, 64\}$ $\gamma_0 \in \{0.05, 0.1, 0.16, 0.25, 0.5, 0.8\}$	$(16, 0.1)$
Full-batch SFRAMES (c_β, γ_0)	$c_\beta \in \{0.25, 1, 4, 16, 64\}$ $\gamma_0 \in \{0.05, 0.1, 0.16, 0.25, 0.5, 0.8\}$	$(16, 0.1)$
SPRING–SARAH, (a, q)	(0.7,16), (1.1,16), (1.6,16), (1.1,8), (1.1,32)	(1.6,16)
ProxSARAH, (a, m)	(0.9,12), (1.5,12), (2.3,12), (1.5,6), (1.5,24)	(2.3,12)
PALM, a	0.6, 1.1, 1.6, 2.3	2.3

We jointly select (c_β, γ_0) for each SFRAMES variant by the smallest median repaired-to-initial training-loss ratio over the two development slices and both training-view counts at 8,640 views per direction. When a selected pair lies on a grid edge, that dimension is extended while retaining the full Cartesian product. Table 2 gives the final tested grids, and Figure 3 shows their observed scores. Held-out views and reference images do not enter selection. The initial calibration fixed $c_\beta = 1$ by this curvature-scaling heuristic and varied only γ_0. All SFRAMES evaluation results reported here use the joint selection. The SPRING–SARAH, ProxSARAH and PALM step multipliers are at the upper edges of their original grids.

Evaluation permits 17,280 forward and adjoint views each. Counters include all gradient evaluations, refreshes, snapshots and both terms of recursive differences. A gradient call returns a joint or block gradient. Each iteration updates both factors. Data processed is $W = \sum_{\ell=1}^{G} |B_\ell|$, counting the views in each of the G gradient calls. One processed view uses one forward and one adjoint over all 956 bins. Diagnostic forward projections are evaluated separately. Evaluation records the initialization and every optimizer update, and tables use each run's last iterate within budget. Paired percentage changes are computed within each seed pair, then summarized by medians over seeds within slices and across slices.

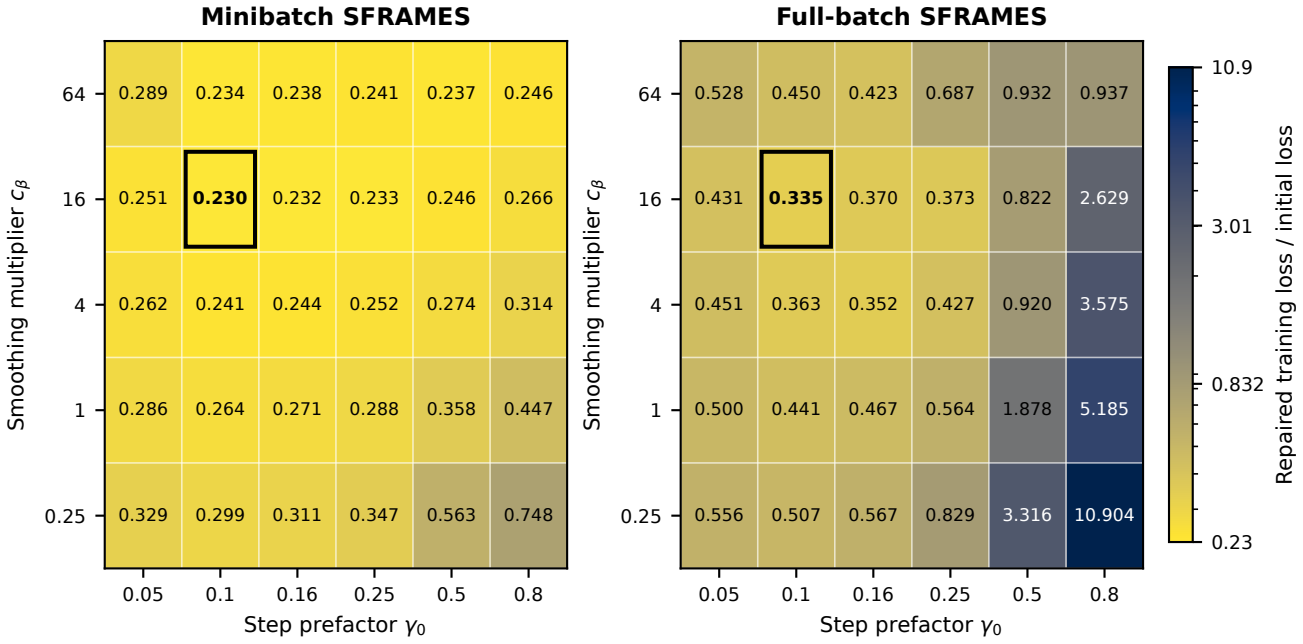


Figure 3: Joint development search. Each cell gives the median repaired training loss relative to initialization across the two development slices and both training-view counts. Here $\beta_0 = c_\beta / \max(L_U, L_V)$. Boxes mark the selected pairs, and lower values are better. The color scale is shared and logarithmic.

References

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